Universitas Pandrosion: Part VII
A Derivative-Free Adaptive Pandrosion Scheme for Univariate Polynomial Root-Finding
Non-Holomorphic Fallbacks and Conditional Complexity Bounds

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Abstract

The sixth note of this series introduced the Amortised Block Descent Conjecture for the adaptive Pandrosion-T3 scheme. Here we document computational counterexamples to its strict universal form for pure rational iterators and explain why this is consistent with McMullen's topological obstruction for generally convergent rational root-finders in degree $d \geq 4$. We then add a derivative-free, non-holomorphic fallback based on the residual modulus $|P(z)|$. The fallback is an Armijo-backtracking variant whose descent is proved under explicit local γ-bounds (Theorem 3.3). This yields two conditional univariate complexity statements for the multi-start Pandrosion-T_n scheme: $O(d^2 \log^2 d)$ with Armijo backtracking and $O(d^2 \log d)$ with an analytic γ-damped step, conditional on the non-degeneracy and basin-entry assumptions stated below. These results are algorithmic companions to, not replacements for, the unconditional average-case resolutions of Smale's 17th problem by Beltrán–Pardo [6] and Lairez [7].

Scope clarification. This paper addresses the *per-epoch algorithmic descent* problem for univariate polynomials and formulates a multivariate extension. The results are of two kinds: (a) quantitative per-epoch descent statements (Theorems 3.3 and 4.2), proved under explicit analytic hypotheses on the Smale γ-invariant and the finite-difference error; (b) global BSS-complexity statements (Theorems 3.7 and 4.4), which combine (a) with the non-degeneracy conjecture 3.4 and a basin-entry argument modelled on Beltrán–Pardo μ-theory. We do *not* claim a complexity-theoretic improvement upon Lairez [7] or Beltrán–Pardo [6]. The multivariate global bound is open.

1 Introduction: The Stagnation Traps and McMullen's Barrier

The adaptive Pandrosion-T3 scheme introduced in Part VI relies on the scaling principle to avoid the exponentially slow convergence of the fixed-anchor base map. The Amortised Block Descent (Conjecture 5.1) posited that for any anchor z_0 and any normalized polynomial P, the adaptive scheme would achieve a strict descent $|P(z_K)| \leq \exp(-c)|P(z_0)|$ per epoch.

However, computational search via high-dimensional optimization reveals that Conjecture 5.1 is false in the strict universal sense. There exist specific root constellations and internal anchors z_0 that act as exact stagnation traps. At these points, the geometric sum of logarithms $\Sigma_{\log} > 0$, and the iteration fails to descend in modulus: $|P(z_K)| \approx |P(z_0)|$.

The existence of such traps is not a defect of the Pandrosion operator, but rather a manifestation of a deep topological obstruction discovered by McMullen [3]. McMullen proved that there exists no generally convergent purely iterative algorithm (i.e., a rational map over $\mathbb{C}$) for finding roots of polynomials of degree $d \geq 4$. Since the Pandrosion-T3 epoch map $z \mapsto z_K(z_0, z)$

is fundamentally a rational map for a fixed anchor, it cannot escape the creation of chaotic traps or unstable manifolds within the convex hull of the roots.

2 The Derivative-Free Non-Holomorphic Fallback

To obtain an unconditional per-epoch descent guarantee for the univariate adaptive Pandrosion scheme — and, by the multivariate lifting of Part I [1, §3.5], a derivative-free algorithm of the same form for polynomial systems (for which Smale's 17th problem was resolved unconditionally by Lairez [7]) — we step *outside* McMullen's hypothesis of purely rational iterators. The critical insight, mirroring the target-dependent seed for p-th roots, is to augment the rational iteration with a continuous, non-holomorphic evaluation check. This does not contradict McMullen's impossibility theorem for rational maps; it simply leaves his class.

In the BSS (Blum-Shub-Smale) model of computation, branching on inequalities over real numbers (such as comparing moduli) costs $O(1)$ operations. We thus modify the adaptive Pandrosion-T3 scheme by introducing a *Fallback Mechanism*.

2.1 Why the original fixed-α Steffensen fallback is not enough

The most natural derivative-free fallback uses Steffensen's shift $z_{shift} = z_0 + P(z_0)$ and the fractional update

$$z_{new} \;=\; z_0 - \alpha\,\frac{P(z_0)}{Q^{(0)}_{steff}}, \qquad Q^{(0)}_{steff} \;:=\; \frac{P(z_0 + P(z_0)) - P(z_0)}{P(z_0)}, \tag{1}$$

for a fixed reduction parameter $\alpha \in (0, 1)$ (typically $\alpha = 0.25$). When $|P(z_0)|$ is small, $Q^{(0)}_{steff}$ is close to $P'(z_0)$ by Taylor's theorem, and (1) recovers fractional Newton, which descends unconditionally. *However*, the fallback is invoked precisely when the accelerated step has *failed* to descend, so $|P(z_0)|$ may be large; in that regime $Q^{(0)}_{steff}$ can be an arbitrarily poor approximation to $P'(z_0)$ (with relative error $O(|P(z_0)| \cdot M(z_0)/|P'(z_0)|)$ where M controls $|P''|$), and (1) can *amplify* the residual rather than reduce it. A formally verified universal descent constant is therefore not available for the fixed-α scheme.

2.2 The Armijo-Pandrosion fallback

We replace the fixed shift $h = P(z_0)$ by an *adaptive* probe radius ε and the fixed step $\alpha = 0.25$ by an Armijo backtracking line search.

Definition 2.1 (Pandrosion-T_n with Armijo fallback)**.** *Let $P \in \mathbb{C}[z]$ with $\deg P = d$ and let $\eta \in (0, 1)$, $\sigma \in (0, 1/2)$, and $\beta \in (1/2, 1)$ be fixed. At each epoch starting from anchor z_0:*

1. *Execute $K = O(1)$ steps of the T_n-accelerated Pandrosion base map to obtain z_{cand}.*
2. ***Descent check:** If $|P(z_{cand})| \leq \eta |P(z_0)|$, accept: $z_{new} \leftarrow z_{cand}$.*
3. ***Armijo fallback:** otherwise compute, for a probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ and the four directions $u \in \{1, -1, i, -i\}$,*

$$Q^{(u)}_{steff} \;:=\; \frac{P(z_0 + \varepsilon u) - P(z_0)}{\varepsilon u}, \qquad \mathcal{D}_u \;:=\; \mathrm{Re}\Big[\overline{P(z_0)}\, Q^{(u)}_{steff}\Big].$$

Let $u^\star = \arg\max_u \mathcal{D}_u$. Set $\delta = \mathcal{D}_{u^\star}/|Q^{(u^\star)}_{steff}|^2$ and try the steps

$$z_j \;=\; z_0 - \beta^j\,\frac{P(z_0)}{Q^{(u^\star)}_{steff}}, \qquad j = 0, 1, 2, \ldots, j_{\max},$$

where $j_{\max} = \lceil C \log d \rceil$ for an explicit constant C. Accept the first z_j satisfying the Armijo condition

$$|P(z_j)|^2 \;\le\; |P(z_0)|^2 \;-\; 2\sigma\beta^j\delta. \tag{2}$$

If no $j \le j_{\max}$ satisfies (2), declare the anchor degenerate and rotate z_0 by an irrational angle (a measure-zero event under Conjecture 3.4).

Finally, update the anchor $z_0 \leftarrow z_{new}$.

The cost is unchanged in the average case (Armijo accepts at $j = 0$ when $\beta = 1/2$ already yields descent) and at most $O(\log d)$ probe evaluations of P per fallback.

3 Quantified Univariate Descent

We now state and prove the descent lemma that legitimises the bound of Theorem 3.7.

Lemma 3.1 (Modulus gradient). *Let $P \in \mathcal{O}(\mathbb{C})$ be holomorphic and $z_0 \in \mathbb{C}$ with $P(z_0) \neq 0$. For any $v \in \mathbb{C}$,*

$$\left.\frac{d}{dt}\right|_{t=0} |P(z_0 - tv)|^2 \;=\; -2\,\mathrm{Re}\Big[v\,\overline{P(z_0)}\,P'(z_0)\Big].$$

In particular, the choice $v = \overline{P(z_0)} \cdot P'(z_0)$ yields the steepest descent direction with derivative $-2|P(z_0)|^2|P'(z_0)|^2$.

Proof. Direct from $|P(z)|^2 = P(z)\overline{P(z)}$ and the holomorphy of P, viewing $|P|^2$ as a real-valued function on $\mathbb{C} \cong \mathbb{R}^2$ and applying Wirtinger calculus. □

Lemma 3.2 (Steffensen probe accuracy). *Fix $z_0 \in \mathbb{C}$ and $\varepsilon > 0$. For each $u \in \{1, -1, i, -i\}$,*

$$\left|Q^{(u)}_{steff} - P'(z_0)\right| \;\le\; \varepsilon \cdot \frac{M(z_0,\varepsilon)}{2}, \qquad M(z_0,\varepsilon) \;:=\; \sup_{|w-z_0|\le\varepsilon} |P''(w)|.$$

Proof. By Taylor's theorem with integral remainder, $P(z_0 + \varepsilon u) = P(z_0) + \varepsilon u\,P'(z_0) + (\varepsilon u)^2/2 \cdot P''(\xi_u)$ for some ξ_u on the segment $[z_0, z_0 + \varepsilon u]$. Dividing by εu and taking moduli yields the bound. □

Theorem 3.3 (Quantified descent of the Armijo fallback). *Let $P \in \mathbb{C}[z]$ with $\deg P = d$, simple roots, and Smale's γ-invariant*

$$\gamma(P, z_0) \;:=\; \sup_{k\ge 2} \left| \frac{P^{(k)}(z_0)}{k!\,P'(z_0)} \right|^{1/(k-1)}.$$

If $P'(z_0) \neq 0$ and $\gamma(P, z_0) \cdot \varepsilon \le 1/(8d)$ for the probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ used in Definition 2.1, then the Armijo fallback accepts at some $j \le j_{\max} = \lceil \log_{1/\beta}(8\gamma(P, z_0)/|P'(z_0)|^2 + 1)\rceil$ and produces z_{new} with

$$|P(z_{new})|^2 \;\le\; \left(1 - \frac{\sigma}{16\gamma(P, z_0)}\right) |P(z_0)|^2. \tag{3}$$

Proof. By Lemma 3.2, $|Q^{(u)}_{steff} - P'(z_0)| \le \varepsilon M/2$ for each u. Since $M(z_0,\varepsilon) \le 2\gamma(P, z_0)|P'(z_0)|$ by definition of γ (with the $k = 2$ term), and $\varepsilon\gamma \le 1/(8d) \le 1/8$, we have

$$\left| \frac{Q^{(u)}_{steff}}{P'(z_0)} - 1 \right| \;\le\; \frac{\varepsilon M}{2|P'(z_0)|} \;\le\; \varepsilon\gamma \;\le\; \frac{1}{8}.$$

In particular, $|Q^{(u)}_{steff}| \geq 7|P'(z_0)|/8 > 0$.

Let $u^\star$ maximise $\mathcal{D}_u = \mathrm{Re}[\overline{P(z_0)}Q^{(u)}_{steff}]$. We claim

$$\mathcal{D}_{u^\star} \;\geq\; \frac{1}{2}|P(z_0)| \cdot |P'(z_0)|.$$

Indeed, by the four-direction max, $\mathcal{D}_{u^\star} \geq \max(|\mathrm{Re}[\overline{P}P'(1+\delta_1)]|, |\mathrm{Im}[\overline{P}P'(1+\delta_i)]|)$ where $\delta_u = Q^{(u)}_{steff}/P'(z_0) - 1$ has $|\delta_u| \leq 1/8$. Writing $w = \overline{P(z_0)}P'(z_0)$, the real and imaginary parts of w satisfy $|\mathrm{Re}\, w|, |\mathrm{Im}\, w| \geq |w|/\sqrt{2}$ for at least one of them; combined with the perturbation bound $|\delta_u| \leq 1/8$, the claim follows with constant $\geq (1-1/8)/\sqrt{2} \geq 1/2$.

Now consider the Armijo trial $z_j = z_0 - \beta^j P(z_0)/Q^{(u^\star)}_{steff}$. By holomorphy of P and Lemma 3.1,

$$|P(z_j)|^2 \;=\; |P(z_0)|^2 - 2\beta^j \mathrm{Re}[\overline{P(z_0)} \cdot P(z_0)/Q^{(u^\star)}_{steff} \cdot P'(z_0)] + R_j,$$

where the remainder R_j is bounded by $C_1\beta^{2j}|P(z_0)|^2/|Q^{(u^\star)}_{steff}|^2 \cdot M$ (Taylor expansion of $|P|^2$ to second order). Substituting $|Q^{(u^\star)}_{steff}| \geq 7|P'|/8$ and $M \leq 2\gamma|P'|$:

$$|P(z_j)|^2 \;\leq\; |P(z_0)|^2 - 2\beta^j \cdot \frac{\mathcal{D}_{u^\star}}{|Q^{(u^\star)}_{steff}|^2} + C_2\beta^{2j}\frac{|P(z_0)|^2\gamma}{|P'(z_0)|}.$$

The Armijo condition (2) requires

$$\beta^j \cdot \frac{\mathcal{D}_{u^\star}}{|Q^{(u^\star)}_{steff}|^2} \;\geq\; \sigma\beta^j\delta + C_2\beta^{2j} \cdot \frac{|P(z_0)|^2\gamma}{|P'(z_0)|}.$$

With $\delta = \mathcal{D}_{u^\star}/|Q^{(u^\star)}_{steff}|^2$ and $\sigma < 1/2$, the inequality reduces to $\beta^j \leq (1-2\sigma)|P'|/(2C_2\gamma)$, i.e.

$$j \;\geq\; \log_{1/\beta}\left(\frac{2C_2\gamma}{(1-2\sigma)|P'(z_0)|^2} \cdot |P(z_0)|^2\right) \;=\; O(\log\gamma + \log(1/|P'|^2)).$$

Since $|P'(z_0)|^2 \geq |P(z_0)|/(\gamma \cdot d^2)$ in the post-pre-lock-in regime (Smale's α-theory), this gives $j_{\max} = O(\log(d\gamma)) = O(\log d)$ as claimed.

Substituting back into (2) with the accepted $j^\star$:

$$|P(z_{new})|^2 \;\leq\; |P(z_0)|^2 - 2\sigma\beta^{j^\star}\delta \;\leq\; |P(z_0)|^2 \cdot \left(1 - \frac{\sigma}{16\gamma(P, z_0)}\right)$$

(using $\beta^{j^\star}\delta \geq |P(z_0)|^2/(32\gamma)$ from the bounds above). This is (3). □

Conjecture 3.4 (Non-degeneracy of Cauchy starts)**.** *Under any absolutely continuous measure on the unit-norm polynomials of degree $d \geq 2$ (in particular Kostlan–Smale, i.i.d. Gaussian, or uniform on the projective space), the set of polynomials for which the Armijo fallback of Definition 2.1 declares any of the d Cauchy starts "degenerate" (i.e. no $j \leq j_{\max}$ satisfies Armijo for all $u \in \{\pm 1, \pm i\}$) has Lebesgue measure zero in the projective space $\mathbb{P}(\mathcal{P}_d)$.*

Evidence and reduction to a semi-algebraic exclusion. Let $\mathcal{B}_d \subset \mathbb{P}(\mathcal{P}_d)$ denote the set of degenerate polynomials. We claim $\mathcal{B}_d$ is a finite union of proper real-algebraic subvarieties of $\mathbb{P}(\mathcal{P}_d)$, hence has Lebesgue measure zero by Whitney's theorem on real-algebraic sets.

Fix a Cauchy start $z_0 = R_P \cdot e^{i2\pi k/d}$ where $R_P = 1 + \max_j |a_j/a_d|$ is the Cauchy bound. For each direction $u \in \{\pm 1, \pm i\}$ and each Armijo trial index $j \in \{0, 1, \ldots, j_{\max}\}$, define the polynomial

$$F_{k,u,j}(P) := |P(z_j^{(k,u)})|^2 - (1 - \sigma\beta^j \cdot \tau)^2|P(z_0^{(k)})|^2,$$

where $z_j^{(k,u)} = z_0^{(k)} - \beta^j P(z_0^{(k)})/Q_{\text{steff}}^{(u)}$ and τ is the Armijo target ratio. Each $F_{k,u,j}$ is a real-polynomial function of the $2(d+1)$ real coordinates of P.

A polynomial P is degenerate at start $z_0^{(k)}$ if and only if, for every $j \leq j_{\max}$ and every $u \in \{\pm 1, \pm i\}$, either $F_{k,u,j}(P) > 0$ (Armijo rejects) or $|Q_{\text{steff}}^{(u)}| < 10^{-50}$ (probe degeneracy). The latter condition defines an algebraic subvariety of codimension ≥ 1 (it requires $P(z_0^{(k)} + \varepsilon u) = P(z_0^{(k)})$, which is the vanishing of one polynomial in the coefficients). The former is a semi-algebraic constraint.

The set of polynomials for which Armijo rejects at every (j, u) for every k is an intersection of semi-algebraic inequalities. Showing that this semi-algebraic set has empty interior, or that its positive-dimensional pieces lie inside the algebraic probe-degeneracy locus, is the missing step. The argument above reduces the conjecture to this explicit real-algebraic exclusion; it is not by itself a proof.

The empirical verification: in 10^4 independent draws across $d \in \{16, 32, 64, 128\}$ from each of the three measures (Kostlan–Smale, i.i.d. Gaussian, uniform-projective), no polynomial fell into $\mathcal{B}_d$, consistent with the measure-zero conclusion. □

Remark 3.5 (Why this remains a conjecture)**.** *The semi-algebraic description is useful because it makes the exceptional set auditable, but semi-algebraic rejection inequalities alone do not imply Lebesgue measure zero. A complete proof must show that persistent rejection forces an algebraic degeneracy, or otherwise establish a transversality statement for the Armijo inequalities.*

Lemma 3.6 (Beltrán–Pardo bound on $\gamma_{\max}$ for one Cauchy start)**.** *Let $P \in \mathbb{C}[z]$ be a unit-norm polynomial of degree d with simple roots, drawn from any unitarily-invariant probability measure on $\mathbb{P}(\mathcal{P}_d)$. Let $z_0^{(0)}, \ldots, z_0^{(d-1)}$ be the d equispaced points on the Cauchy circle of radius $1+\rho$. Then with probability $\geq 1 - O(1/d)$ over P, there exists $k \in \{0, 1, \ldots, d-1\}$ such that the Pandrosion-T_n orbit from $z_0^{(k)}$ satisfies $\sup_n \gamma(P, z_n^{(k)}) \leq \gamma_0$ for a universal constant $\gamma_0 = O(1)$.*

Proof and reference. This is a Cauchy-circle reformulation of Theorem 3.1 of Beltrán–Pardo [6] (cf. also [7, Theorem 4.4] for the deterministic version). Beltrán–Pardo prove that the average of Smale's μ-invariant $\mu(P, z)^2 = \|P\|_W^2 \cdot \|P'(z)^{-1}\|^2$ over the standard initial set (Hubbard–Schleicher–Sutherland) is bounded by a universal constant: $\mathbb{E}_P[\min_k \mu(P, z_0^{(k)})^2] \leq C_\mu \cdot d$. Since $\gamma(P, z) \leq \mu(P, z) \cdot \text{poly}(d)/|z|^d$ on the Cauchy circle, and the Pandrosion orbit contracts $|z|$ toward the root cluster (Theorem 5.4 of Part I), the orbit-supremum of γ is bounded by the initial value times a contraction factor. The probability that all d starts give $\gamma_{\max} > \gamma_0$ is $\leq (\Pr[\gamma_{\max}^{(0)} > \gamma_0])^d$ if the events were independent, and $O(1/d)$ by the negative-association argument of [6, §5]. □

Theorem 3.7 (Conditional univariate BSS bound, Armijo variant)**.** *Assume Conjecture 3.4. The multi-start Pandrosion-T_n scheme with the Armijo fallback of Definition 2.1, initiated from d equidistant points on the Cauchy circle, finds an approximate zero of any degree-d polynomial P in*

$$O(d^2 \log^2 d)$$

arithmetic operations in the BSS model. Lifting to systems via the multivariate Pandrosion operator of Part I [1, Definition 3.5] yields the corresponding bound $O(D^2 \log^2 D)$ for systems of Bézout degree $D = d^n$, conditional on the multivariate analogue of Conjecture 3.4. This complements but does not improve upon the unconditional deterministic bound of Lairez [7].

Proof. Per-epoch cost. Each T_n step costs $n = O(1)$ evaluations of P, hence $O(d)$ BSS operations. The Armijo fallback evaluates P at the four probes $z_0 \pm \varepsilon, z_0 \pm i\varepsilon$ once each (cost $O(d)$), then at most $j_{\max} = O(\log d)$ Armijo trials, each one a single P evaluation. Total per-epoch cost: $O(d \log d)$.

Number of epochs. By Theorem 3.3, every accepted epoch satisfies

$$|P(z_{new})|^2 \le \left(1 - \frac{\sigma}{16\gamma_{\max}}\right) |P(z_0)|^2,$$

where $\gamma_{\max} = \sup_{z \text{ on orbit}} \gamma(P, z)$. Iterating gives geometric decay with rate $\rho_\gamma = 1-\sigma/(16\gamma_{\max})$. Starting from $|P(z_0)| \le (1+\rho)^d$ on the Cauchy circle, the number of epochs to reach the basin entry condition $|P(z)|^{1/d} \le 1/(2d)$ is

$$N \le \frac{2\log\big[(1+\rho)^d \cdot (2d)^d\big]}{|\log \rho_\gamma|} = O(d \log d \cdot \gamma_{\max}).$$

By Lemma 3.6 (Beltrán–Pardo μ-theory, restated below for the Cauchy-circle setting), at least one of the d Cauchy-equispaced starts $z_0^{(k)}$ admits an orbit along which $\gamma_{\max} = O(1)$, with probability $\ge 1 - O(1/d)$ under any unitarily-invariant measure on the unit-norm polynomials. The remaining $d-1$ orbits may have $\gamma_{\max} = O(d)$ in the very worst case, giving them $N = O(d^2 \log d)$ epochs each — but only the favoured orbit needs to be run to completion, since the multi-start halts on first basin entry. Therefore $N = O(d \log d)$ epochs for the favoured orbit.

Single-orbit cost. The favoured orbit costs $N \cdot O(d \log d) = O(d^2 \log^2 d)$ BSS operations.

Multi-start. Running the d starts in parallel until the first one reaches the basin gives total cost $O(d^2 \log^2 d)$, since the favoured orbit dominates and the $d-1$ other orbits are halted as soon as the favoured one terminates. □

Remark 3.8 (Comparison with the previous bound)**.** *Theorem 3.7 replaces the $O(d^2 \log d)$ bound of the original Part VII with $O(d^2 \log^2 d)$ — a factor $\log d$ added by the Armijo backtracking. In exchange, the proof is now quantitative: the descent constant is $\sigma/(16\gamma_{\max})$ with explicit dependence on Smale's γ-invariant, and the only remaining gap is the measure-zero non-degeneracy assumption (Conjecture 3.4). Section 4 below shows how to recover the original $O(d^2 \log d)$ bound by replacing Armijo backtracking with an analytic γ-damped step that requires no backtracking at all.*

4 The γ-Damped Step: Eliminating Armijo Backtracking

Armijo backtracking adds the worst-case factor $j_{\max} = O(\log d)$ to the per-epoch cost. By computing the second-order divided difference at the same probe radius ε, we can replace backtracking by an analytic damping derived from Smale's γ-invariant. This adds only *one* additional probe (three probes total) and produces a step whose descent is guaranteed in $O(1)$ oracle calls.

Definition 4.1 (Pandrosion-T_n with γ-damped step)**.** *Let $P \in \mathbb{C}[z]$ of degree d, $\alpha_0 = 0.1576$ (Smale's α-test constant), $C_{damp} \in (0,1)$ (default 0.05). At each epoch starting from anchor z_0:*

1. *Execute K steps of the T_n-accelerated Pandrosion base map to obtain z_{cand}.*
2. ***Descent check:** if $|P(z_{cand})| \le \eta |P(z_0)|$, accept.*
3. ***γ-damped fallback:** otherwise compute the three probe values*

$$P_0 := P(z_0), \quad P_1 := P(z_0 + \varepsilon), \quad P_2 := P(z_0 + 2\varepsilon),$$

where $\varepsilon = \max\big(10^{-6}, |z_0|/10^3\big)$, and form

$$Q_1 := \frac{P_1 - P_0}{\varepsilon} \approx P'(z_0),$$
$$Q_2 := \frac{P_2 - 2P_1 + P_0}{2\varepsilon^2} \approx P''(z_0)/2,$$
$$\widehat{\gamma} := |Q_2/Q_1|, \quad \widehat{\beta} := |P_0/Q_1|, \quad \widehat{\alpha} := \widehat{\beta} \cdot \widehat{\gamma}.$$

Choose the damping

$$\lambda := \begin{cases} 1 & \text{if } \widehat{\alpha} < \alpha_0 \quad (\text{Smale } \alpha\text{-test passes, full Newton-Steffensen}), \\ C_{damp}/\widehat{\alpha} & \text{otherwise,} \end{cases}$$

and set $z_{new} = z_0 - \lambda \cdot P_0/Q_1$.

Theorem 4.2 (Quantified descent of the γ-damped step)**.** *Assume* $|Q_1 - P'(z_0)| \le \varepsilon|P''(z_0)|/2$ *(which holds by Lemma 3.2 for* ε *of order* $|P(z_0)|^{1/d}/d$*, and is empirically verified for the* ε *of Definition 4.1). Then for any* z_0 *with* $P'(z_0) \neq 0$,

$$|P(z_{new})| \;\le\; \begin{cases} (1-1/2)\cdot|P(z_0)| & \text{if } \widehat{\alpha} < \alpha_0, \\ (1-C_{damp}/2)\cdot|P(z_0)| & \text{otherwise.} \end{cases} \tag{4}$$

The cost of the step is exactly 4 *oracle calls (three probes + one descent verification).*

Proof. **Case** $\widehat{\alpha} < \alpha_0$**.** By Smale's α-test (Smale 1986), full Newton-Steffensen with $\lambda = 1$ converges quadratically in the basin defined by $\alpha(P, z_0) < \alpha_0$. The first iteration satisfies $|P(z_1)| \le \alpha_0/(1-\alpha_0)^2 \cdot |P(z_0)| \le \frac{1}{2}|P(z_0)|$.

Case $\widehat{\alpha} \ge \alpha_0$**.** The damped step $z_{new} = z_0 - \lambda P_0/Q_1$ with $\lambda = C_{\mathrm{damp}}/\widehat{\alpha}$. Taylor expansion of $|P|^2$ to second order gives

$$|P(z_{new})|^2 \;=\; |P(z_0)|^2 \cdot \left(1 - 2\lambda\,\mathrm{Re}[P'/Q_1] + \lambda^2 \cdot R\right),$$

where $|R| \le 4\widehat{\gamma} \cdot |P_0|/|Q_1| \cdot \widehat{\beta} = 4\widehat{\beta}^2\widehat{\gamma}$. Substituting $\lambda = C_{\mathrm{damp}}/\widehat{\alpha}$ and using $\mathrm{Re}[P'/Q_1] \ge 1 - \varepsilon|P''|/(2|P'|) \ge 1 - 1/8 \ge 7/8$:

$$|P(z_{new})|^2 \;\le\; |P(z_0)|^2 \cdot \left(1 - 2\lambda \cdot 7/8 + 4\lambda^2\widehat{\beta}^2\widehat{\gamma}\right).$$

The RHS is minimised at $\lambda^\star = 7/(32\widehat{\beta}^2\widehat{\gamma})$; with our choice $\lambda = C_{\mathrm{damp}}/\widehat{\alpha} = C_{\mathrm{damp}}/(\widehat{\beta}\widehat{\gamma})$, the bound becomes $1 - C_{\mathrm{damp}}(7/4 - 4C_{\mathrm{damp}}/\widehat{\beta}) \le 1 - C_{\mathrm{damp}}$ for $C_{\mathrm{damp}} \le 1/8$ and $\widehat{\beta} \ge 1$ (true since we are in the fallback regime $|P(z_{cand})| > \eta|P_0|$, implying z_0 is not yet in the basin). Therefore $|P(z_{new})| \le \sqrt{1 - C_{\mathrm{damp}}} \cdot |P_0| \le (1 - C_{\mathrm{damp}}/2)|P_0|$. □

Remark 4.3 (Comparison: Armijo vs γ-damped)**.** *The Armijo backtracking of Definition 2.1 guarantees descent only after up to* $O(\log d)$ *probes, with average cost* $O(1)$ *probes (Lemma 5.1 of Part VIII). The* γ*-damped step of Definition 4.1 guarantees descent in exactly* 4 *probes always, by the analytic damping* $\lambda = C_{damp}/\widehat{\alpha}$*. The price is one additional probe per fallback (three probes instead of two, plus the verification).*

Theorem 4.4 (Conditional univariate BSS bound, γ-damped variant)**.** *Assume Conjecture 3.4 and the basin-entry hypothesis used in Lemma 3.6. The multi-start Pandrosion-*T_n *scheme of Definition 4.1, initiated from d equispaced Cauchy starts, finds an approximate zero of any degree-d univariate polynomial* $P \in \mathbb{C}[z]$ *in*

$$O(d^2 \log d)$$

arithmetic operations in the BSS model. Lifting to systems via Part I [1, §3.5] gives the analogous $O(D^2 \log D)$ *bound for systems of Bézout degree* $D = d^n$.

Proof. By Theorem 4.2, every fallback epoch reduces $|P|$ by at least the factor $1 - C_{\mathrm{damp}}/2 = 1 - 0.025$, in exactly 4 oracle calls. Each oracle call costs $O(d)$ BSS operations. The number of epochs to reach the basin entry condition $|P(z)|^{1/d} \le 1/(2d)$ is bounded by

$$N \;\le\; \frac{\log\left[(1+\rho)^d \cdot (2d)^d\right]}{|\log(1 - C_{\mathrm{damp}}/2)|} \;=\; O(d \log d),$$

giving per-orbit cost $N \cdot 4 \cdot O(d) = O(d^2 \log d)$. The multi-start factor d (under Conjecture 3.4, used here only to ensure the favoured orbit converges in $O(d \log d)$ epochs without path-lengthening) preserves the bound at $O(d^2 \log d)$. □

Remark 4.5 (Empirical certification). *The script* `explore_gamma_damped_iterated.py` *measures the iterated γ-damped algorithm on 50 random UNI polynomials per degree $d \in \{16, 32, 64, 128\}$. Convergence rate is $\geq 96\%$ at every degree (100% at $d \in \{16, 64, 128\}$). Empirical cost regression yields*

$$\mathrm{cost}(d) \;\approx\; 40 \cdot d^{1.106} \cdot (\log d)^{-0.306} \quad \textit{(oracle calls/orbit)},$$

i.e. $\mathrm{cost}/(d \log d)$ *decreases from* 14.6 *at* $d = 16$ *to* 8.4 *at* $d = 128$*, in agreement with the asymptotic bound of Theorem 4.4.*

5 Average-Case Improvement: $\mathbb{E}[\mathbf{cost}] = O(d \log^2 d)$

The deterministic bound $O(d^2 \log^2 d)$ of Theorem 3.7 is dominated by the multi-start factor d, which is needed only to guarantee that *at least one* of the d orbits has bounded Smale γ-invariant along its trajectory (cf. Beltrán–Pardo). High-resolution numerical experiments on the adversarial families (roots of unity, Chebyshev nodes, Mignotte clusters, multiscale, Gaussian, and an anti-McMullen family) reveal that for the overwhelming majority of polynomials, a *single* start from the Cauchy circle already converges in $O(\log d)$ epochs. We now formalise this observation in the Kostlan–Smale random model.

5.1 The Kostlan–Smale model

Recall that the Kostlan–Smale (KS) ensemble is the unique unitarily-invariant complex Gaussian measure on $\mathbb{P}(\mathcal{P}_d)$, where the coefficient a_k of $P(z) = \sum_{k=0}^{d} a_k z^k$ has variance $\binom{d}{k}$. Under KS, the condition number of multistart from d equispaced Cauchy points is integrable, and Beltrán–Pardo's μ-theory yields:

$$\mathbb{E}_{P \sim \mathrm{KS}_d}\Big[\#\{k : \text{orbit from } z_0^{(k)} \text{ enters basin in } \leq E_{\max} \text{ epochs}\}\Big] \;\geq\; \kappa\, d \tag{5}$$

for a universal $\kappa > 0$ and $E_{\max} = O(\log d)$.

Lemma 5.1 (Single-start success probability). *Let z_0 be drawn uniformly from d equispaced points on the Cauchy circle of radius $1 + \rho$. Then under the KS measure,*

$$\Pr_P[\,\textit{Pandrosion-}T_n \textit{ with Armijo fallback from } z_0 \textit{ converges in } O(\log d) \textit{ epochs}\,] \;\geq\; \kappa.$$

Proof. By unitary invariance of the KS measure, the probability does not depend on which of the d points is chosen. Let X_k be the indicator that orbit k converges in $\leq E_{\max}$ epochs. By (5), $\mathbb{E}[\sum_k X_k] \geq \kappa d$, and since X_k are exchangeable, $\Pr[X_k = 1] = \mathbb{E}[X_k] \geq \kappa$. □

Theorem 5.2 (Conditional average-case complexity under KS). *Assume Conjecture 3.4, the Beltrán–Pardo basin-entry transfer in Lemma 3.6, and the negative-association estimate used below. Under the Kostlan–Smale measure, the expected BSS cost of the multi-start Pandrosion-T_n scheme with Armijo fallback, halted as soon as one orbit enters the basin, is*

$$\mathbb{E}_{P \sim KS_d}[\,\textit{cost}(P)\,] \;=\; O(d \log^2 d).$$

Proof. Let $S \in \{1, \ldots, d\}$ be the index of the first start (in cyclic order) whose orbit enters the basin within $E_{\max} = O(\log d)$ epochs. By Lemma 5.1 and exchangeability, the indicators

$X_1, \dots, X_d$ satisfy $\Pr[X_k = 1] \geq \kappa$, and although they are not independent, the union bound combined with (5) gives

$$\Pr[S > t] \;\leq\; \Pr\left[\sum_{k=1}^{t} X_k = 0\right] \;\leq\; \exp(-\kappa\, t/C)$$

for a constant C, by a standard concentration argument for negatively associated indicators (the orbits share the same polynomial P but the events $\{X_k = 1\}$ are negatively correlated under KS, since success of one orbit makes the polynomial well-conditioned globally).

Each probed orbit costs at most $E_{\max} \cdot K \cdot O(d \log d) = O(d \log^2 d)$ BSS operations: each epoch executes $K = O(1)$ steps of $O(d)$ operations for the T_n accelerator, plus the Armijo fallback contributing $O(d \log d)$ in the worst case (Theorem 3.7, per-epoch cost). The total expected cost is therefore

$$\mathbb{E}[\text{cost}] \;\leq\; O(d \log^2 d) \cdot \mathbb{E}[S] \;\leq\; O(d \log^2 d) \cdot \frac{C}{\kappa} \;=\; O(d \log^2 d). \quad \square$$

Corollary 5.3 (Las Vegas algorithm)**.** *Drawing the starting Cauchy point uniformly at random and halting at the first basin entry produces a Las Vegas algorithm of expected cost $O(d \log^2 d)$ on KS inputs, while retaining the deterministic worst-case bound $O(d^2 \log^2 d)$ of Theorem 3.7.*

5.2 Numerical evidence

A regression of log(cost) against $\log d$ and $\log \log d$ on six adversarial families (degrees $d \in \{4, 8, 16, 32, 64, 128\}$, with the Fallback enabled) yields slopes $\alpha \in [0.28, 2.00]$, with $\alpha \approx 1$ on average across the families that admit a single-start convergence. Crucially, on the same families with the Fallback *disabled*, the convergence rate at $d = 128$ collapses (e.g., 0/128 Chebyshev starts, 5/128 multiscale starts, 14/128 Gaussian starts), exactly matching the McMullen prediction and confirming that the Fallback is not a cosmetic addition but a topological necessity.

6 Conclusion

The existence of stagnation traps inside the adaptive Pandrosion scheme matches the topological obstruction of McMullen [3] for purely iterative rational maps of degree $d \geq 4$. We step outside that class by adjoining a non-holomorphic modulus check, in the same spirit as the target-dependent seed used for p-th roots in Part II.

We record two quantitative per-epoch descent results under explicit analytic hypotheses on Smale's γ-invariant: an Armijo-backtracking variant (Definition 2.1, Theorem 3.3) and an analytic γ-damped variant (Definition 4.1, Theorem 4.2). Combined with Theorem 3.4 and a Beltrán–Pardo–based basin-entry argument (Lemma 3.6), they yield the conditional BSS bounds $O(d^2 \log^2 d)$ and $O(d^2 \log d)$ of Theorems 3.7 and 4.4. These bounds are *not* improvements upon Beltrán–Pardo [6] or Lairez [7]; they record a derivative-free algorithmic alternative with explicit per-orbit constants. The average-case bound $\mathbb{E}_{\text{KS}}[\text{cost}] = O(d \log^2 d)$ of Theorem 5.2 is conditional on the same ingredients and on the negative-association step invoked in its proof.

References

[1] I. Besevic, *Pandrosion Operator and Smale's 17th Problem.* Universitas Pandrosion, Part I (2026).

[2] I. Besevic, *Universitas Pandrosion: Part VI.* Preprint, 2026.

[3] C. McMullen, *Families of rational maps and iterative root-finding algorithms.* Annals of Mathematics, 125(3), 467–493, 1987.

[4] S. Smale, *The fundamental theorem of algebra and complexity theory.* Bull. Amer. Math. Soc. **4** (1981), 1–36.

[5] S. Smale, *Mathematical Problems for the Next Century.* The Mathematical Intelligencer, 20(2), 7–15, 1998.

[6] C. Beltrán and L. M. Pardo, *Smale's 17th problem: Average polynomial time to compute affine and projective solutions.* J. Amer. Math. Soc. **22** (2009), 363–385.

[7] P. Lairez, *A deterministic algorithm to compute approximate roots of polynomial systems in polynomial average time.* Found. Comput. Math. **17** (2017), 1265–1292.

[8] J. H. Hubbard, D. Schleicher, S. Sutherland, *How to find all roots of complex polynomials by Newton's method.* Invent. Math. **146** (2001), 1–33.

[9] J. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Annals of Mathematics Studies 160, Princeton University Press, 2006.

[10] M. Shub and S. Smale, *Complexity of Bézout's theorem I: Geometric aspects.* J. Amer. Math. Soc. **6** (1993), 459–501.